OPEN-SOURCE AUDIT REPORT

A Capstone Project for OSS NGMC Course

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INTRODUCTION

Open-source software is deeply embedded in modern computing. Most of the technology we interact with today — operating systems, web servers, programming tools, and everyday applications — is built using open-source methods. Unlike proprietary software, which is owned and controlled by a single company, open-source software is developed by communities of programmers who share their code openly with the world. This collaborative approach drives innovation, encourages transparency, and gives users everywhere the ability to access and adapt software to suit their own needs.

The concept of open source isn't simply about making software free of cost — it's about giving people genuine freedom. The freedom to use it, learn from it, change it, and pass it on. This has built a global culture where developers continuously improve existing tools rather than reinventing the wheel every time. As a result, open-source software has quietly become the backbone of much of today's digital infrastructure

For this project, I chose **VLC Media Player** as the focus of my open-source audit. VLC is a multimedia player developed by the VideoLAN project, capable of playing nearly every audio and video format without requiring any additional codecs. It began as a student project and gradually grew into a powerful, globally recognized tool, shaped by contributions from developers across the world. The core aim of this project is to understand both the philosophy behind open-source software and how it functions practically — through studying VLC's origins and license, examining how it operates within a Linux environment, and writing shell scripts to demonstrate real interaction with the system.

PART A — ORIGIN & PHILOSOPHY

A1. Origin Story — The Problem VLC Was Created to Solve

The story of VLC Media Player is a good example of what open-source development is truly about. It originated in the 1990s at École Centrale Paris, where a group of students were working on a project to stream video across their university network. At the time, the multimedia tools available were either too limited, too rigid, or fully proprietary — meaning users had no ability to modify or adapt them. This made experimenting with video streaming technologies unnecessarily difficult.

That decision proved to be the right one. Over time, VLC grew far beyond its original purpose. What started as a university network project turned into one of the most capable media players ever built. A big part of its success comes from its ability to handle virtually any audio or video format without needing external codecs — making it accessible to users across different platforms and operating systems. Developers worldwide helped refine its performance, fix bugs, add features, and keep it compatible with newer technologies.

The students faced a clear problem: they needed a multimedia solution that was reliable, flexible, and open enough to evolve with their needs. Since nothing like that existed, they built it themselves. This decision wasn't purely technical — it was also a deliberate move away from the restrictions of closed-source software. By releasing VLC as open-source, the developers ensured that others could access the code, improve it, and adapt it freely.

By making VLC source the developers made sure it would keep getting better with help from the community.Over time VLC Media Player became much more than its purpose.What started as a project turned into a powerful multimedia tool.One key reason for its success is that it supports audio and video formats without needing extra codecs.

As a result VLC became one of the trusted and widely used media players globally.The story of VLC Media Player shows how open-source software can solve problems and promote innovation and collaboration.

It proves that even a small project by students can become a recognized tool when built on openness and shared knowledge.VLC Media Player VLC is an example of what can be achieved with open-source development and community contributions.The VLC Media Player has come a way since its beginning and its success is a testament, to the power of open-source software.

A2. License Analysis

VLC Media Player is released under the GNU General Public License (GPL), which is designed to ensure that software remains free in terms of user rights and not just cost. The GPL provides the four essential freedoms of free software: the freedom to use the software for any purpose, the freedom to study how it works by accessing its source code, the freedom to modify it according to personal needs, and the freedom to distribute copies, including modified versions. VLC fully supports all four of these freedoms, which is why it is considered true free and open-source software.

Under the GPL, a company can use and even sell the software, but there is an important condition. If they modify VLC and distribute their version, they must also make the source code available under the same license. This ensures that improvements remain accessible to the community and prevents companies from taking open-source work and turning it into closed proprietary software.

The GPL is different from permissive licenses like MIT. The MIT license allows developers to use, modify, and even convert the software into closed-source products without sharing their changes. In contrast, the GPL enforces openness by requiring all modified versions to remain open source. Personally, I would choose the GPL

license for my own project because it ensures that my work continues to benefit others and cannot be restricted in the future.

There is also an important distinction between “free as in free beer” and “free as in freedom.” VLC is free in both senses: users do not have to pay to use it, and they also have the freedom to modify and share it. This makes VLC a strong example of how licensing can protect both accessibility and user rights.

A3. The Ethics of Open Source

Open source is a lot more than a way to make software. It is a way of thinking about how we share what we know and the tools we use. It is all about being open working together and making progress as a group.. It also makes us think about some big questions.

Not every piece of software has to be source. On the one hand open source is great because it lets developers learn from what other people have done and make it better. It makes technology easier for everyone to use and understand.. There are times when it is better to keep software closed like when we need to protect secret information keep important systems safe or help a company stay ahead of the competition. So both open-source and proprietary models are useful it just depends on the situation.

Another thing to think about is whether it is fair for companies to make money from open-source software that was created by communities. Companies like Red Hat and Google use open-source technologies like Linux a lot. On the one hand it seems fair because these companies often help the community by giving them money developing things and keeping everything running smoothly. On the hand some people might think it is not fair if companies make a lot of money without giving enough back to the people who made the original software.

The idea that we are "standing on the shoulders of giants" is really important in open-source development. It means that developers can use what other people have already done to create things instead of starting from nothing. In this way open source clearly helps us innovate by saving time and getting people to work together.. Some people might think that it can stop us from coming up with completely new ideas if we rely too much on what already exists.

In my opinion open source mostly helps us innovate than holding us back. It creates a world where we can share what we know freely so developers can learn faster. Make better systems. At the time it is really important that we respect the people who contribute to open source and make sure they are supported properly. Open source is all, about source. We need to make sure that open source keeps helping us make new and exciting things.

Part B — Linux Footprint

To understand how VLC Media Player operates within a Linux system, I installed it on an Ubuntu environment and observed how it is managed, where its files are stored, and how it interacts with the system.

Installation Method

VLC Media Player is easily available through the package manager in Ubuntu. It can be installed using the apt package manager, which automatically handles dependencies and updates.

sudo apt update

sudo apt install vlc

This shows that VLC is well-integrated into the Linux ecosystem and does not require manual compilation from source for normal usage.

Key Directories

After installation, VLC files are distributed across different directories in the system:

- /usr/bin/vlc → This is the main executable file used to launch VLC
- /usr/lib/ → Contains libraries and supporting files required for VLC
- /etc/ → Stores system-wide configuration files (if applicable)
- ~/.config/vlc → Contains user-specific configuration and preferences
- /var/log/ → Used for system logs (VLC itself does not generate extensive logs by default)

These directories show how Linux organizes software into separate components such as binaries, configurations, and user data.

User and Group

VLC Media Player runs under the currently logged-in user rather than a dedicated system user. For example, if the logged-in user is rajkumar, VLC will run with the same user permissions.

This is important for security because it prevents VLC from accessing or modifying system-level files without permission. Running applications as a normal user reduces the risk of system damage in case of vulnerabilities or errors.

Service Management

Unlike server-based software, VLC is not typically managed as a background service. It is an application that is started manually by the user.

It can be launched using:

vlc

or

cvlc # command-line version

Although VLC is not controlled using system services like systemctl, it can still be used in scripts or automated workflows if required.

Update Model

VLC updates are managed through the Linux package management system. When the open-source community releases updates or security patches, they are distributed through official repositories.

Users can update VLC using:

sudo apt update

sudo apt upgrade

This ensures that users receive the latest features, bug fixes, and security improvements without manually downloading updates.

Overall, VLC Media Player integrates smoothly into the Linux system. Its installation, file structure, and update process reflect the standard practices followed by open-source software. This makes it easy to manage, secure, and maintain within a Linux environment.

Part C. The FOSS Ecosystem

Open source software is part of a group of tools and communities. VLC Media Player shows us how different open source parts work together to make an application.

Dependencies and Supporting Tools

VLC Media Player needs open source libraries and tools to work properly. One important tool is FFmpeg. It helps with encoding and decoding audio and video. VLC Media Player also uses codec libraries to support multimedia formats. These tools help VLC Media Player play any media file. The user does not need to install anything

VLC Media Player also uses standard Linux system libraries and development tools like GCC. It uses these tools for compilation and other runtime libraries that support interfaces and system interactions.

Influence and Contributions

VLC Media Player has influenced other multimedia applications and tools. It can handle formats and streaming. This has inspired open source media players and streaming solutions. VLC Media Player has contributed a lot to the development of multimedia standards and cross platform media support.

VLC Media Player has also helped users and developers try out streaming, media conversion and broadcasting. This makes it useful for more than playing media.

Relation to the Web and LAMP Stack

VLC Media Player is not directly part of the LAMP stack. The LAMP stack is used for web development.. Vlc Media Player plays a role in web related applications through media streaming and content delivery.

For example VLC Media Player can stream video content over a network. This content can then be accessed through web applications. This helps the web ecosystem by enabling multimedia integration.

Community and Governance

The VideoLAN organization. Maintains VLC Media Player. They manage the development and coordinate contributions from developers around the world. The project has a community. This community contributes through forums, mailing lists and online repositories.

Developers and users can report bugs suggest features and contribute code. This helps make sure the software keeps getting better and will be around, for a time.

Overall

VLC Media Player is connected to the open source ecosystem. It depends on open source technologies. It contributes to multimedia development.. It is supported by a global community. This shows how open source projects grow and evolve through collaboration and shared effort. VLC Media Player is an example of this. It shows how open source software can be powerful and flexible.

Part D — Open Source vs Proprietary (Critical Analysis)

To understand the real value of VLC Media Player, it is important to compare it with a proprietary alternative such as Windows Media Player or other commercial media players. This comparison highlights both strengths and limitations of open-source software.

Comparison Table

Dimension	VLC (Open Source)	Proprietary Media Player
Cost	Completely free to use and distribute	Often comes pre-installed or requires paid features
Security	Source code is open and can be audited by anyone	Source code is closed and controlled by the company
Support and Reliability	Community support through forums and documentation	Official support from the company
Freedom to Modify	Users can modify and customize the software	No modification allowed
Community vs Control	Developed by a global community (VideoLAN)	Controlled by a single company

Analysis and Verdict

From this comparison, it is clear that VLC offers a high level of freedom, transparency, and flexibility. Users are not restricted by licensing limitations and can modify or distribute the software freely. The open-source nature also improves security, as anyone can review the code and identify potential issues. In addition, VLC supports a wide range of formats without requiring additional purchases, making it highly practical for everyday use.

However, proprietary software also has certain advantages. It often provides dedicated customer support, polished user interfaces, and tighter integration with specific operating systems. In some cases, companies may offer better reliability guarantees or professional support services, which can be important for enterprise environments.

Overall, I would recommend VLC Media Player for real-world use, especially for individual users and general multimedia needs. It is reliable, flexible, and cost-effective. I would also consider contributing to such an open-source project in the future, as it promotes learning, collaboration, and continuous improvement within the developer community.

Shell Script Tasks

Script 1 System Identity Report

Script:

```
#!/bin/bash
# Script 1: System Identity Report
# Author: SRIDHAR SB | Course: Open Source Software #
--- Variables ---

STUDENT_NAME="SRIDHAR"
SOFTWARE_CHOICE="VLC Media Player"
# --- System info ---
KERNEL=$(uname -r)
USER_NAME=$(whoami)

USER_HOME=$HOME
UPTIME=$(uptime -p)

DATE_TIME=$(date '+%d %B %Y, %I:%M:%S %p')
# Get Linux distribution

if [ -f /etc/os-release ];
then    . /etc/os-release

   DISTRO=$PRETTY_NAME
else

   DISTRO="Unknown Linux Distribution"
fi
# License message

LICENSE_MSG="Linux is built on open-source principles. The
kernel is licensed under GPLv2, ensuring freedom to use,
modify, and share."


# --- Display ---

echo "================================================="
```

```
echo "        Open Source System Report"
echo "==================================================="
echo "Student Name      : $STUDENT_NAME"
echo "Chosen Software   : $SOFTWARE_CHOICE"
echo "---------------------------------------------------"
echo "Operating System : $DISTRO"
echo "Kernel Version    : $KERNEL"
echo "Logged-in User    : $USER_NAME"
echo "Home Directory    : $USER_HOME"
echo "Current Date      : $DATE_TIME"
echo "System Uptime     : $UPTIME"
echo "---------------------------------------------------"
echo "About Open Source:"
echo "$LICENSE_MSG"
echo "==================================================="
```

Output:

```
===================================================
 Open Source System Report
===================================================
Student Name : SRIDHAR
Chosen Software : VLC Media Player
---------------------------------------------------
Operating System : Ubuntu 24.04.4 LTS
Kernel Version : 6.6.87.2-microsoft-standard-WSL2
Logged-in User : geekx
Home Directory : /home/geekx
Current Date : 31 March 2026, 05:40:03 PM
System Uptime : up 0 minutes
---------------------------------------------------
About Open Source:
Linux is built on open-source principles. The
kernel is licensed under GPLv2, ensuring freedom to use,
modify, and share.
===================================================
```

Explanation:

This script displays basic system information such as Linux distribution, kernel version, current user, uptime, and date/time. It uses variables to store system values and command substitution to fetch real-time data. The script demonstrates the use of echo for formatted output and helps understand how Linux system details can be accessed through shell scripting.

Script 2 FOSS Package Inspector

Script:

```
#!/bin/bash

# Script 2: FOSS Package Inspector
# Author: SRIDHAR SB (24MEI10077)


# --- Input ---

PACKAGE=${1:-"vlc"} # Default package = VLC


echo "==================================================="
echo "      FOSS Package Inspection Report"

echo "==================================================="
echo "Target Package : $PACKAGE"

echo "---------------------------------------------------"


# --- Check if package is installed ---
if dpkg -s "$PACKAGE" &>/dev/null; then

   echo "Status       : Installed ✓"


   # Fetch details

   VERSION=$(dpkg-query -W -f='${Version}\n' "$PACKAGE")

   SUMMARY=$(dpkg-query -W -f='${Description}\n' "$PACKAGE" |
head -1)



   echo "Version       : $VERSION"
echo "Description     : $SUMMARY" else


   echo "Status       : Not Installed " fi
```

```
echo "--------------------------------------------------"

echo "Open Source Insight:"


# --- Philosophy section ---

case "$PACKAGE" in

    vlc)

        echo "VLC Media Player: Started as a student project
in Paris, VLC became a global tool that plays almost any media
format without restrictive licensing."

        ;;

    apache2|httpd)

        echo "Apache: The backbone of the web, showing how
community collaboration can power most of the internet."

        ;;

    mysql|mysql-server)

        echo "MySQL: A widely used database that highlights
how open source can coexist with commercial models."

        ;;

    firefox)

        echo "Firefox: A browser built to protect openness,
privacy, and user control on the web."

        ;;

    git)

        echo "Git: Created by Linus Torvalds, it
revolutionized version control and decentralized
collaboration."

        ;;

    *)

        echo "$PACKAGE: A part of the open-source ecosystem
contributing to shared knowledge and innovation."

        ;;

esac


echo "==================================================="
```

Output:

```
==============================================
        FOSS Package Inspection Report
==============================================
Target Package   : vlc
----------------------------------------------
Status           : Installed ✔
Version          : 3.0.20-3build6
Description      : multimedia player and streamer
----------------------------------------------
Open Source Insight:
VLC Media Player: Started as a student project in Paris, VLC became a global tool that plays almost any media format wit
hout restrictive licensing.
==============================================
rajkumar_s@RajKumar:~$
```

Explanation:

This script checks if VLC Media Player is on your computer and shows information about it. I made it take a package name. It uses `vlc` by default. It uses the `dpkg` command to see if its installed.

If VLC is installed the script gets the version and a short summary of the package using `dpkg-query`. It shows this info. If VLC is not installed it says the package is not there. You can see this in the output. The script. Shows details or a "not installed" message.

The script has a `case` part. It shares a thought about VLC Media Player. VLC is a free multimedia tool. It plays any media format. This shows how important VLC is, in the open-source world.

This script is an example. It uses `if-statements, command substitution and `case` statements. It shows how shell scripting can check software on your computer. It gives you info based on your computers state.

Script 3 Disk Auditor

Script:

```
#!/bin/bash

# Script 3: Disk and Permission Auditor
# Author: SRIDHAR SB (24MEI10077)


# --- Directories to audit ---
DIRS=("/etc"     "/var/log"     "/home"     "/usr/bin"     "/tmp"
"$HOME/.config/vlc")


echo "==================================================="
echo "      Disk and Permission Audit Report"
echo "==================================================="


echo "This report shows directory permissions, ownership, and
disk usage."
echo "---------------------------------------------------"


# --- Loop through directories ---
for DIR in "${DIRS[@]}"; do
   if [ -d "$DIR" ]; then


     # Get permissions, owner, and group

     PERMS=$(ls  -ld  "$DIR"  |  awk  '{print  $1}')
OWNER=$(ls   -ld   "$DIR"   |   awk   '{print   $3}')
GROUP=$(ls -ld "$DIR" | awk '{print $4}')


     # Get size (ignore permission errors)
     SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)
```

```
        # Display nicely formatted output
        printf "%-25s | %s | Owner: %s:%s | Size: %s\n" \
        "$DIR" "$PERMS" "$OWNER" "$GROUP" "${SIZE:-N/A}"

    else
        printf "%-25s | Not Found \n" "$DIR"
    fi
done

echo "-------------------------------------------------"
echo "VLC Configuration Check:"

# --- VLC specific check ---
if [ -d "$HOME/.config/vlc" ]; then
    echo "VLC configuration folder is present."
    echo "This folder stores user preferences and settings in
open, editable format."
else
    echo "VLC configuration folder not found."
    echo "Try running VLC once to generate user configuration
files."
fi

echo "=================================================="
```

Output:

```
==================================================
        Disk and Permission Audit Report
==================================================
This report shows directory permissions, ownership, and disk usage.
--------------------------------------------------
/etc                      | drwxr-xr-x | Owner: root:root | Size: 4.4M
/var/log                  | drwxrwxr-x | Owner: root:syslog | Size: 208M
/home                     | drwxr-xr-x | Owner: root:root | Size: 104K
/usr/bin                  | drwxr-xr-x | Owner: root:root | Size: 98M
/tmp                      | drwxrwxrwt | Owner: root:root | Size: 20K
/home/rajkumar_s/.config/vlc | drwxr-xr-x | Owner: rajkumar_s:rajkumar_s | Size: 8.0K
--------------------------------------------------
VLC Configuration Check:
VLC configuration folder is present.
This folder stores user preferences and settings in open, editable format.
==================================================
rajkumar_s@RajKumar:~$
```

Explanation:

This script is designed to analyze important system directories and display their permissions, ownership, and disk usage. In my implementation, I created an array of directories such as /etc, /var/log, /home, /usr/bin, /tmp, and the VLC configuration folder located at ~/.config/vlc.

The script uses a for loop to iterate through each directory in the list. For every directory, it first checks whether it exists using an if condition. If the directory is present, the script uses the ls -ld command along with awk to extract permissions, owner, and group information. It also uses the du -sh command to calculate the size of the directory while suppressing permission errors.

The output is formatted using printf to display the directory name along with its permissions, ownership, and size in a clean and readable manner. If a directory does not exist, the script prints a message indicating that it is not found on the system.

Additionally, the script includes a specific check for the VLC configuration folder. If VLC has been used on the system, this folder will exist and contain user preferences. Otherwise, the script informs the user that the folder is not present, which is also reflected in the output.

Script 4 — Log Analyzer

Script:

```
#!/bin/bash

# Script 4: Log File Analyzer

# Author: SRIDHAR SB (24MEI10077) | Course: Open Source
Software


LOGFILE=$1

KEYWORD=${2:-"error"} # Default keyword = error
COUNT=0

echo "==================================================="
echo "       Log File Analysis Report"

echo "==================================================="


# --- Validate input file (do-while style loop) ---
while true; do

   if [ -z "$LOGFILE" ]; then

     read -p "Enter log file path: " LOGFILE
continue

   fi


   if [ ! -f "$LOGFILE" ]; then

     echo "File not found: $LOGFILE"

     read -p "Try again (or type 'exit'): " LOGFILE
[ "$LOGFILE" = "exit" ] && exit 1

     continue
fi


   if [ ! -s "$LOGFILE" ]; then

     echo "File is empty: $LOGFILE"
```

```
        read -p "Enter a non-empty file (or type 'exit'): "
LOGFILE
        [ "$LOGFILE" = "exit" ] && exit 1
        continue
    fi

    break
done

echo "Analyzing file   : $LOGFILE"
echo "Searching for    : '$KEYWORD'"
echo "--------------------------------------------------"
# --- Count keyword occurrences ---
while IFS= read -r LINE; do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"

echo "Total Matches    : $COUNT"
echo "--------------------------------------------------"
# --- Display last matches ---
if [ "$COUNT" -gt 0 ]; then
    echo "Last 5 matching entries:"
    grep -i "$KEYWORD" "$LOGFILE" | tail -n 5
else
    echo "No matching entries found."
fi

echo "==================================================="
```

Output:

```
==================================================
           Log File Analysis Report
==================================================
Enter log file path: /var/log/syslog
Analyzing file    : /var/log/syslog
Searching for     : 'error'
--------------------------------------------------
Total Matches     : 45
--------------------------------------------------
Last 5 matching entries:
grep: /var/log/syslog: binary file matches
==================================================
rajkumar_s@RajKumar:~$ |
```

Explanation:

This script helps us look at a log file and see how times a certain word is in it. When we use the script it asks us to type in the path to the log file we want to look at. This path is what we see when the script says "Enter log file path:". This makes the script easy to use with log files. When we give the script the file path it checks to make sure the file is really there and that it is not empty. If the file is not there or if it is empty the script shows us an error message. Asks us to try again. This helps the script work properly even if we make a mistake.

The script then looks at the log file one line at a time. For each line it checks to see if the word we are looking for is in that line. By default the script looks for the word "error". If it finds the word it adds one to a counter. After it looks at the file the script tells us how many times the word is in the log file.

The script also shows us the few lines where it found the word. This helps us see what is going on with the system. If there are any problems. We can use this to fix things that are not working right.

This script is an example of how we can use loops, conditional statements and text processing in Bash. It is a useful tool for looking at log files and figuring out what is going on. We can use it to help us debug and monitor the system. The script is about log files and how to analyze them. Log files are important, for debugging and monitoring. The script helps us with log files.

Script 5 — Manifesto Generator
Script:

```
#!/bin/bash

# Script 5: Open Source Manifesto Generator #
Author: SRIDHAR SB (24MEI10077)


# --- Alias (concept demonstration) ---
shopt -s expand_aliases

alias header='echo "========== Open Source Manifesto
=========="'


echo "==================================================="
echo "      Open Source Manifesto Generator"

echo "==================================================="
echo "Answer the following questions to generate your
manifesto."

echo "---------------------------------------------------"


# --- User Input ---

read -p "1. Name one open-source tool you use daily: " TOOL
read -p "2. In one word, what does 'freedom' mean to you? "
FREEDOM

read -p "3. What would you build and share openly? " BUILD



# --- System Info --- DATE=$(date
'+%d %B %Y') USER_NAME=$(whoami)
OUTPUT_FILE="manifesto_${USER_NAME}.txt"



echo "---------------------------------------------------"
echo "Generating your personalized manifesto..."
```

```
echo "--------------------------------------------------"

# --- Write to file ---
header > "$OUTPUT_FILE"

echo "" >> "$OUTPUT_FILE"
echo "Date: $DATE" >> "$OUTPUT_FILE"
echo "Author: $USER_NAME" >> "$OUTPUT_FILE"
echo "" >> "$OUTPUT_FILE"

echo "On this day, I reaffirm my belief in the open-source
philosophy." >> "$OUTPUT_FILE"
echo "Tools like $TOOL remind me that collaboration drives
innovation." >> "$OUTPUT_FILE"
echo "To me, freedom means '$FREEDOM'." >> "$OUTPUT_FILE"
echo "I aspire to build and share $BUILD so that others can
learn, adapt, and grow." >> "$OUTPUT_FILE"
echo "Together, we build upon each other's work and move
technology forward." >> "$OUTPUT_FILE"

# --- Display Output ---
echo ""
echo "Manifesto successfully saved to: $OUTPUT_FILE"
echo "--------------------------------------------------"
echo "Here is your manifesto:"
echo "--------------------------------------------------"

cat "$OUTPUT_FILE"

echo "==================================================="
```

Output:

```
==================================================
        Open Source Manifesto Generator
==================================================
Answer the following questions to generate your manifesto.
--------------------------------------------------
1. Name one open-source tool you use daily: VLC Player
2. In one word, what does 'freedom' mean to you? sharing
3. What would you build and share openly? AI Tools
--------------------------------------------------
Generating your personalized manifesto...
--------------------------------------------------

Manifesto successfully saved to: manifesto_rajkumar_s.txt
--------------------------------------------------
Here is your manifesto:
--------------------------------------------------
========== Open Source Manifesto ==========

Date: 31 March 2026
Author: rajkumar_s

On this day, I reaffirm my belief in the open-source philosophy.
Tools like VLC Player remind me that collaboration drives innovation.
To me, freedom means 'sharing'.
I aspire to build and share AI Tools so that others can learn, adapt, and grow.
Together, we build upon each other's work and move technology forward.
==================================================
rajkumar_s@RajKumar:~$
```

Explanation:

This script generates a personalized open-source manifesto based on user input. In my implementation, the script first displays a message asking the user to answer three questions. It then uses the read command to take inputs such as the open-source tool used, the meaning of freedom, and what the user wants to build and share. These inputs are stored in variables and later used to construct a meaningful paragraph.

The script also captures the current date and the system username using commands like date and whoami. It then creates an output file named based on the user and writes the generated manifesto into it using file redirection (> and >>).

In the output, the user can see both the generated manifesto and the file where it is saved. This script demonstrates user interaction, string handling, and file operations in Bash in a practical and meaningful way.

Conclusion:

This project gave me an understanding of open-source software. I learned about it by studying it and actually doing it. I chose VLC Media Player to see how a real open-source tool is made, given out and kept up to date on Linux. I found out how a simple project made by a student became a known application all around the world because people worked together and shared their work.

I learned a lot about licenses and ethics. I realized that open source is not just about getting software for free. It is about people being free to use it it being transparent and it being made by a community. This also made me think about what's fair who contributes and what companies do in the open-source world.

Working on Linux and making shell scripts really helped me get better at the side of things. Each script showed me something like looking at the system managing packages setting file permissions looking at logs and talking to users. These tasks showed me how automation can make system work easier and more efficient.

From a technical perspective, working on Linux and developing shell scripts improved my practical skills significantly. Each script demonstrated different concepts such as system inspection, package management, file permissions, log analysis, and user interaction. These tasks showed how automation can be used to simplify and analyze system operations effectively.

Overall this project showed me how important open-source software is in computing today. It showed me that when people work together and share what they know they can make flexible tools. From this experience I got skills and a better understanding of what open-source development is all about which will help me in my future work, as a developer.